Fiscal Year:	FY 2009	**Task Last Updated:**	FY 07/01/2009
PI Name:	Baulch, Janet E. Ph.D.		
Project Title:	Dose rate effects on the mechanisms of space radiation induced delayed genomic instability		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Irvine	**State:**	CA
Zip Code:	92697	**Congressional District:**	45
Comments:	NOTE: PI previously at University of Maryland; moved to UCA Irvine in summer 2013 (Ed., 11/11/13)		
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
Start Date:	09/01/2007	**End Date:**	08/31/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2011 (from 8/31/2010) per S. Krenek/JSC (7/2010)		
Key Personnel Changes/Previous PI:	No Changes		
COI Name (Institution):	Morgan, William (University of Maryland)		
Grant/Contract No.:	NNX07AT42G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Those brave individuals traveling in space are exposed to a variety of radiations. It is known that exposure to ionizing radiation can lead to a greater risk for developing cancer later. Since there is currently no way to protect astronauts from space radiations, it is important that the biological consequences of exposures to these radiations are understood. Using proton and iron ion radiations we will investigate the delayed effects of radiation exposure based on radiation quality. Specifically, we will investigate those deleterious effects observed in the progeny of irradiated cells multiple cell generations after the initial exposure. These deleterious effects are grouped under the term of radiation-induced genomic instability and are thought to be an early step in the carcinogenic process. The primary objective of this proposal is to evaluate the differences in dose rate effects and radiation quality among x-rays, protons, and iron ions in the induction of delayed genomic instability and epigenetic aberrations. We will accomplish this by measuring cell survival, gene mutations/deletions and rearrangements, and other forms of chromosome damage. Further this project will explore the		

	possibility that DNA methylation or miRNA mediated gene silencing is involved in the radiation response and radiation-induced genomic instability.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Some of the research and technology questions that this proposal addresses include the following: 1) What are the probabilities for increased carcinogenesis from space radiation as a function of NASA's operational parameters (age at exposure, age, latency, gender, tissue, mission, radiation quality, dose rate, and exposure protraction)? 2) How can effective biomarkers of carcinogenic risk from space radiation be developed and validated? 3) What is the likelihood that DNA damage will increase the risk of carcinogenesis? We hypothesize that radiation quality affect the induction of delayed genomic instability and aberrant DNA methylation (thought to be cancer precursors) following exposure to space radiations. We will test the hypothesis that radiation quality affects the induction of delayed genomic instability following exposure to radiations commonly encountered in space, specifically iron ions and protons. Using a panel of assays (e.g. chromosomal instability, micronuclei (MN) formation, delayed mutation, and reproductive integrity) we will compare the biological effectiveness of protons, iron ions and x-rays for the induction of delayed genomic instability. We hypothesize that altered DNA methylation is an epigenetic effect of exposure to space radiations that has a mechanistic role in genomic instability and is a measure of carcinogenic risk. We will provide a greater understanding of the potential relationship between radiation-induced genomic changes and epigenetic factors based on radiation quality. We will develop and validate high throughput biomarkers for genomic instability endpoints and DNA methylation. These hypotheses will be tested by evaluating global, repeat element, and specific locus DNA methylation at immediate and delayed times post-irradiation. Among the loci evaluated for changes in methylation in this study are p16 and MGMT. This promoter hypermethylation and gene silencing has been linked to a number of specific cancers and is thought to be a biomarker for increased carcinogenic risk. These methylation assays are PCR based and will provide a high-throughput assay for the assessment of carcinogenic risk.
Task Progress:	Some of the research and technology questions that this proposal addresses include the following: 1) What are the probabilities for increased carcinogenesis from space radiation as a function of NASA's operational parameters (age at exposure, age, latency, gender, tissue, mission, radiation quality, dose rate, and exposure protraction)? 2) How can effective biomarkers of carcinogenic risk from space radiation be developed and validated? 3) What is the likelihood that DNA damage will increase the risk of carcinogenesis? We hypothesize that radiation quality affects the induction of delayed genomic instability and aberrant DNA methylation (thought to be cancer precursors) following exposure to space radiations. Specific Aim 1 will test the hypothesis that dose rate and radiation quality affect the induction of delayed genomic instability following exposure to radiations commonly encountered in space, specifically iron ions and protons. Using a panel of assays (e.g. chromosomal instability, micronuclei (MN) formation, delayed mutation, and reproductive integrity) we will compare the biological effectiveness of dose rate effects and radiation quality among protons, low LET x-ray radiation and high LET heavy ion radiation for the induction of delayed genomic instability. We hypothesize that altered DNA methylation is an epigenetic effect of exposure to space radiations that has a mechanistic role in genomic instability and is a measure of carcinogenic risk. We will provide a greater understanding of the potential relationship between radiation-induced genomic changes and epigenetic factors. We will develop and validate high throughput biomarkers for genomic instability endpoints and DNA methylation. These hypotheses will be tested by evaluating global, repeat element and specific locus DNA methylation at immediate and delayed times post-irradiation following exposure to protons, iron ions and x-rays. Among the loci evaluated for changes in methylation in this study are p16 and MGMT for whom inappropriate promoter methylation leads to gene silencing. This promoter hypermethylation and gene silencing has been linked to a number of specific cancers, and is thought to be a biomarker for increased carcinogenic risk. These methylation assays are PCR based and will provide a high-throughput assay for the assessment of carcinogenic risk. Accomplishments for year 2: 1. Development and validation of three DNA methylation assays has been completed. a. Abritrarily priming methylation sensitive PCR (AP-MSP) b. Repeat element methylation assays for LINE-1 and alu elements c. Methylation sensitive specific locus assays for p16 and MGMT promoter methylation 2. Application of these DNA methylation assays to evaluate the relative effect of iron ions, protons or x-rays on DNA methylation. Data from this study shows increased genome wide methylation, decreased repeat element methylation, and no effect on p16 and MGMT promoter methylation. Little difference is observed between the effect of iron ions and protons on these endpoints. These epigenetic effects occur at radiation doses for which no detectible effect on DNA damage as measured by micronucleus induction or cell killing in clonogenic cell survival assays. X-ray data is still being collected. 3. Application of DNA methylation assays to evaluate cell lines that exhibit radiation induced genomic instability. Using the AP-MSP, we found clear differences in genomic methylation between stable and all unstable clones. Changes in repeat element methylation were observed for some unstable clones, but not consistently for all of them. These results suggested that aberrant DNA methylation is associated with genomic instability, but that it may not be at specific sequences within the genome. We have also generated miRNA profiles of the stable and unstable clones. Again, there are clear differences in miRNA profiles between stable and unstable cell lines. The miRNA data is now being mined for specific miRNAs that may manipulate pathways of interest for follow-up studies.
Bibliography Type:	Description: (Last Updated: 05/01/2017)

Abstracts for Journals and Proceedings	Aypar U, Morgan WF, Baulch JE. "Epigenetic Alterations in Radiation Induced Genomic Instability." 54th Annual Radiation Research Society meeting, Boston, MA, September 21-25, 2008. 54th Annual Radiation Research Society meeting, Boston, MA, September 21-25, 2008. , Sep-2008
Abstracts for Journals and Proceedings	Aypar U, Malkiel LN, Baulch JE. "Effects of high and low LET irradiation on DNA methylation and genomic instability." High LET/space radiation effects of the 54th Annual Radiation Research Society meeting, Boston, MA, September 21-25, 2008. 54th Annual Radiation Research Society meeting, Boston, MA, September 21-25, 2008. , Sep-2008
Abstracts for Journals and Proceedings	Aypar U, Morgan WF, Baulch JE. "Epigenetic Alterations in High and Low LET Radiation Induced Genomic Instability." Space radiation effects at the 39th Annual Environmental Mutagen Society meeting, Rio Grande, Puerto Rico, October 18-22, 2008. 39th Annual Environmental Mutagen Society meeting, Rio Grande, Puerto Rico, October 18-22, 2008. , Oct-2008
Abstracts for Journals and Proceedings	Aypar U, Malkiel NM, Baulch JE. "Effects of Irradiation on DNA Methylation and Genomic Instability." Space radiation risk at the 39th Annual Environmental Mutagen Society meeting, Rio Grande, Puerto Rico, October 18-22, 2008. 39th Annual Environmental Mutagen Society meeting, Rio Grande, Puerto Rico, October 18-22, 2008. , Oct-2008
Articles in Peer-reviewed Journals	Sowa MB, Goetz W, Baulch J, Pyles D, Dziegielewski J, Yovino S, Snyder AR, Targanox de Toledo S, Azzam E, Morgan W. "Lack of evidence for low-LET radiation induced bystander response in normal human fibroblasts and colon carcinoma cells." Submitted to International Journal of Radiation Biology, March 2009. , Mar-2009